



Book I. Prop. XXXV. Theor. *Parallelograms on the same base, and between the same parallels, are (in area) equal.*

On account of the parallels,

$$\left. \begin{array}{l} \text{Red triangle} = \text{Blue triangle}; \\ \text{Black triangle} = \text{White triangle}; \end{array} \right\} \text{(pr. I.29, I.34).}$$

and $\text{Blue triangle} = \text{Red triangle}$

But $\text{Yellow triangle} = \text{Yellow triangle}$ (pr. I.8)

$$\therefore \text{Yellow triangle} - \text{Blue triangle} = \text{Yellow triangle} - \text{Red triangle},$$

$$\text{and } \text{Yellow triangle} - \text{Blue triangle} = \text{Yellow triangle} - \text{Red triangle};$$

$$\therefore \text{Yellow triangle} = \text{Yellow triangle}.$$

Q. E. D.